

Prasuna Podhigari

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SUMMARY

Software Engineer with 5 years of experience designing, developing, testing, deploying, and supporting production applications, backend services, data-processing systems, and cloud infrastructure across financial-services and technology environments. Strong hands-on experience with Java, Python, React, TypeScript, SQL, Spring Boot, FastAPI, PostgreSQL, Kafka, AWS, Docker, Kubernetes, and Terraform. Experienced building REST APIs, responsive web applications, distributed and event-driven systems, data-processing workflows, and secure cloud services. Comfortable across the complete SDLC, including system design, development, automated testing, code review, CI/CD, release management, production troubleshooting, application resiliency, and incident response. Experienced using AI-assisted engineering tools for coding, refactoring, testing, debugging, and documentation while validating generated output for correctness, performance, security, and maintainability.

EXPERIENCE

Flash Parking, United States | Associate Software Engineer

October 2024 – July 2026

- Built and maintained Java Spring Boot microservices and REST/GraphQL APIs supporting reservation, payment, customer-account, search, and parking-availability workflows using Spring MVC, Spring Security, Spring Data JPA, Hibernate, and PostgreSQL.
- Developed customer-facing applications using React, TypeScript, Redux, HTML, and CSS, building reusable components for parking search, reservations, payments, accounts, location-based workflows, and operational interfaces.
- Led the migration of customer-facing parking functionality from Mapbox to Google Maps, rebuilding shared map, marker, navigation, search, bounds, and location components across multiple repositories while coordinating with product, UX, QA, and backend teams.
- Built Python and FastAPI services for data processing, payment validation, reservation lookup, customer-support automation, and integrations with internal Java services and third-party APIs.
- Developed Python and SQL workflows to process reservation, transaction, payment, customer, and operational data, performing validation, transformation, cleansing, aggregation, and reconciliation before exposing processed data through APIs and internal reporting workflows.
- Wrote and optimized PostgreSQL queries using joins, CTEs, aggregations, indexes, transaction handling, and execution-plan analysis to support application workflows, production troubleshooting, reporting, and high-volume transaction processing.
- Built event-driven and asynchronous systems using Kafka and RabbitMQ for reservation events, payment processing, inventory updates, notifications, retries, and communication across Java, Python, and Go services.
- Developed Go services for parking search, availability, inventory synchronization, scheduled jobs, payment gateways, and third-party integrations using goroutines, channels, REST, and gRPC.
- Designed and maintained AWS-backed services across EC2, ECS, Lambda, API Gateway, S3, CloudFront, IAM, DynamoDB, and related cloud services, supporting both application and data-processing workloads.
- Provisioned cloud infrastructure through Terraform and CloudFormation and deployed containerized services using Docker and Kubernetes across development and production environments.
- Maintained CI/CD pipelines using GitHub Actions, Jenkins, Maven, and Git, supporting automated builds, testing, security checks, packaging, deployment, rollback, and post-release validation.
- Used AI-assisted development tools including GitHub Copilot, Claude, and Gemini for code exploration, refactoring, unit-test generation, debugging, and technical documentation while validating generated output through peer review, automated testing, secure coding standards, and production engineering practices.
- Applied responsible AI practices by avoiding sensitive production information in AI prompts and independently validating AI-generated code for correctness, performance, security, resiliency, and maintainability before integration.
- Supported production applications using CloudWatch, Grafana, Graylog, Rollbar, PostgreSQL, and OpenSearch, troubleshooting API failures, slow queries, application errors, deployment issues, messaging failures, and distributed-service incidents.
- Participated in production on-call support, delivering hotfixes, root-cause analysis, incident resolution, post-deployment validation, and corrective actions for customer-impacting issues.
- Worked across requirements, technical design, implementation, testing, code review, deployment, and operational support in an Agile environment while mentoring junior engineers and collaborating with product, UX, QA, platform, and operations teams.

Ally Financial, United States | Software Developer

January 2024 – September 2024

- Developed Java Spring Boot microservices and REST APIs for customer accounts, transaction processing, dispute workflows, document management, reporting, and operational applications using Spring MVC, Spring Security, JPA, Hibernate, and PostgreSQL.
- Built interactive and responsive React and TypeScript applications for customer information, transaction review, document tracking, case management, search, and operational dashboards.
- Integrated React applications with Spring Boot APIs using asynchronous request handling, reusable API services, application state management, form validation, error handling, and secure authentication through OAuth2 and JWT.
- Developed Python services and automation scripts for transaction validation, data processing, document analysis, reconciliation, report generation, and operational workflow automation.
- Wrote and tuned complex SQL queries for data retrieval, manipulation, validation, reconciliation, and analytical use cases across customer, transaction, dispute, and workflow datasets.
- Designed PostgreSQL schemas, indexes, relationships, and transaction models while investigating query performance and application data issues in production environments.
- Built asynchronous integrations using Kafka to process transaction events, workflow updates, notifications, and downstream application messages.
- Developed Python data-processing pipelines to ingest information from REST APIs, relational databases, and JSON payloads, applying

transformation and automated cleansing before downstream consumption.

- Worked closely with product managers, analysts, operations teams, and business stakeholders to translate business requirements into technical specifications, API contracts, SQL logic, application workflows, and reporting functionality.
- Containerized backend applications using Docker, deployed services through Kubernetes, and supported AWS infrastructure including EC2, S3, Lambda, API Gateway, IAM, RDS, and related services.
- Provisioned infrastructure through Terraform and CloudFormation and supported Jenkins and GitHub Actions pipelines for automated testing, deployment, and release validation.
- Used AI-assisted engineering tools for code generation, refactoring, unit tests, troubleshooting, and documentation, reviewing generated output for functional correctness, security, data sensitivity, and maintainability before adoption.
- Participated in the complete SDLC including requirements analysis, design, development, testing, peer review, deployment, production monitoring, incident resolution, and Agile ceremonies.

DSquare Tech, India | Software Developer

January 2021 – December 2022

- Developed enterprise applications using Java, Spring Boot, JavaScript, SQL, and PostgreSQL for digital payment processing, transaction validation, customer verification, merchant management, and fraud-detection workflows.
- Built REST APIs and backend services using Spring MVC, Spring Data JPA, Hibernate, validation, exception handling, and relational data models.
- Developed frontend components using Angular, React, JavaScript, HTML, and CSS for transaction review, merchant operations, customer verification, payment status, and internal operational workflows.
- Built reusable Angular components and services, integrated REST APIs, handled forms, validation, routing, and basic state management for internal payment and customer-service applications.
- Designed PostgreSQL tables, indexes, joins, stored queries, and transactional logic supporting high-volume payment, customer, merchant, and fraud-review data.
- Implemented Kafka producers and consumers for payment authorization, transaction status, customer verification, fraud-scoring, and asynchronous downstream processing.
- Built Python scripts for transaction validation, data extraction, reconciliation, log processing, and recurring operational reports. Supported Maven and Jenkins-based CI/CD pipelines, Git workflows, automated testing, application deployments, and production releases.
- Developed automated tests using JUnit and Mockito, validated REST APIs with Postman, and participated in code reviews, debugging, defect resolution, and regression testing.
- Investigated production issues using application logs, SQL queries, API responses, and transaction records, working with development, QA, infrastructure, and business teams to identify root causes and deliver fixes.
- Participated in Agile/Scrum development and supported applications throughout requirements, development, testing, deployment, and production support.

PROJECTS

AgentProtocol – A2A Multi-Agent System - GitHub

Python | LangChain | Agentic AI | Tool Calling | REST APIs

- Built an A2A-style multi-agent architecture using Python and LangChain, with independently scoped Weather and Math agents communicating through tool-based delegation and asynchronous agent execution.
- Implemented agent boundaries, system prompts, task routing, tool invocation, context transfer, response validation, and fallback behavior to demonstrate interoperable specialized agents collaborating on multi-step requests.

Production Data Processing & Operations Platform

Python | FastAPI | PostgreSQL | Kafka | AWS | Docker | Terraform

- Built a Python/FastAPI service that ingested high-volume application and transaction events through REST APIs and Kafka, validated and transformed incoming data, persisted normalized records in PostgreSQL, and exposed processed results through internal REST endpoints.
- Added asynchronous processing, retry handling, data-quality validation, structured logging, and operational health endpoints, then containerized the service with Docker and provisioned AWS infrastructure using Terraform.
- Built SQL-based operational reports to identify failed processing jobs, transaction anomalies, latency patterns, and recurring service issues, providing engineers and operations teams with actionable production insights.

SKILLS

Core Technologies: Java, Python, TypeScript, JavaScript, SQL, Spring Boot, FastAPI, React, Redux, REST APIs, Microservices, Distributed Systems

Platform: PostgreSQL, DynamoDB, Redis, Kafka, RabbitMQ, AWS, Docker, Kubernetes, Terraform, CloudFormation, CI/CD

AI-Assisted Engineering: GitHub Copilot, Claude, Gemini, OpenAI, Code Generation, Refactoring, Test Generation, Troubleshooting, Documentation

Engineering Practices: Data Structures & Algorithms, Object-Oriented Programming, System Design, API Design, Secure Coding, Automated Testing, Code Reviews, Agile Development, Application Resiliency, Debugging, Production Support

EDUCATION

Master of Science in Computer Science | University of Central Missouri, Missouri, USA

January 2023 – May 2024

CERTIFICATIONS

AWS Cloud Practitioner

Claude Certified Architect, Foundations